Report: DHCP Scope Configuration and Lease Process in Packet Tracer

Tool: Cisco Packet Tracer | **Topic:** DHCP server, address pool, client leasing, release and renew

1. Objective

To configure a DHCP scope on a server, connect clients that obtain their addresses automatically, and document how the lease process works, including release and renew, the lease table, and the four-step DORA exchange.

2. Topology and Addressing

A DHCP server and three client PCs connect to one switch on a single 192.168.1.0/24 network.

```
PC0 ----\\
PC1 ----- Switch ---- DHCP Server (192.168.1.2)
PC2 ----/
```

The screenshots confirm the server address (192.168.1.2) and the client settings handed out. The pool values below follow the lab plan; check them against your server's DHCP service page.

Item	Value
Network	192.168.1.0/24
DHCP server (static)	192.168.1.2
Default gateway given to clients	192.168.1.1
DNS server given to clients	192.168.1.2
Pool start address	192.168.1.10
Subnet mask	255.255.255.0
Maximum users	50

The server and gateway addresses sit below the pool start, so DHCP never hands them out.

3. Configuration

Server (Server > Services > DHCP):

- Service: On
- Pool name: OfficePool
- Default gateway: 192.168.1.1
- DNS server: 192.168.1.2
- Start IP address: 192.168.1.10
- Subnet mask: 255.255.255.0
- Maximum number of users: 50
- Click Save so the pool appears in the table

The server itself needs a static address (Desktop > IP Configuration): 192.168.1.2, mask 255.255.255.0, since a DHCP server cannot get its own address from DHCP.

Clients (Desktop > IP Configuration): select **DHCP** on each PC. No addresses are typed in by hand.

4. How the Lease Process Works (DORA)

When a client sets itself to DHCP, it goes through four steps to get an address:

Step	Packet	Sent by	What happens
D	Discover	Client (broadcast)	The client has no address, so it shouts to the whole LAN: any DHCP server out there?
O	Offer	Server	The server picks a free address from the pool and offers it, with mask, gateway and DNS
R	Request	Client	The client accepts and formally asks for the offered address
A	Acknowledge	Server	The server confirms the lease and records it

The Discover and Request are broadcasts because the client does not yet have an IP address. After the Acknowledge, the client configures itself with the address and keeps it for the lease time.

5. Test Results

5.1 Release: returning the lease

On PC2, a ping to 192.168.1.10 first succeeded with 0% loss (reply TTL 128, so the host is on the same subnet with no router in between). Running `ipconfig /release` then handed the lease back: the IP address, subnet mask, default gateway and DNS server all fell to 0.0.0.0. The `ipconfig /all` output afterward shows the IPv4 address as 0.0.0.0, while the DHCP Servers field still remembers 192.168.1.2 as the server that issued the old lease. PC2's MAC address is 0005.5E65.7975.

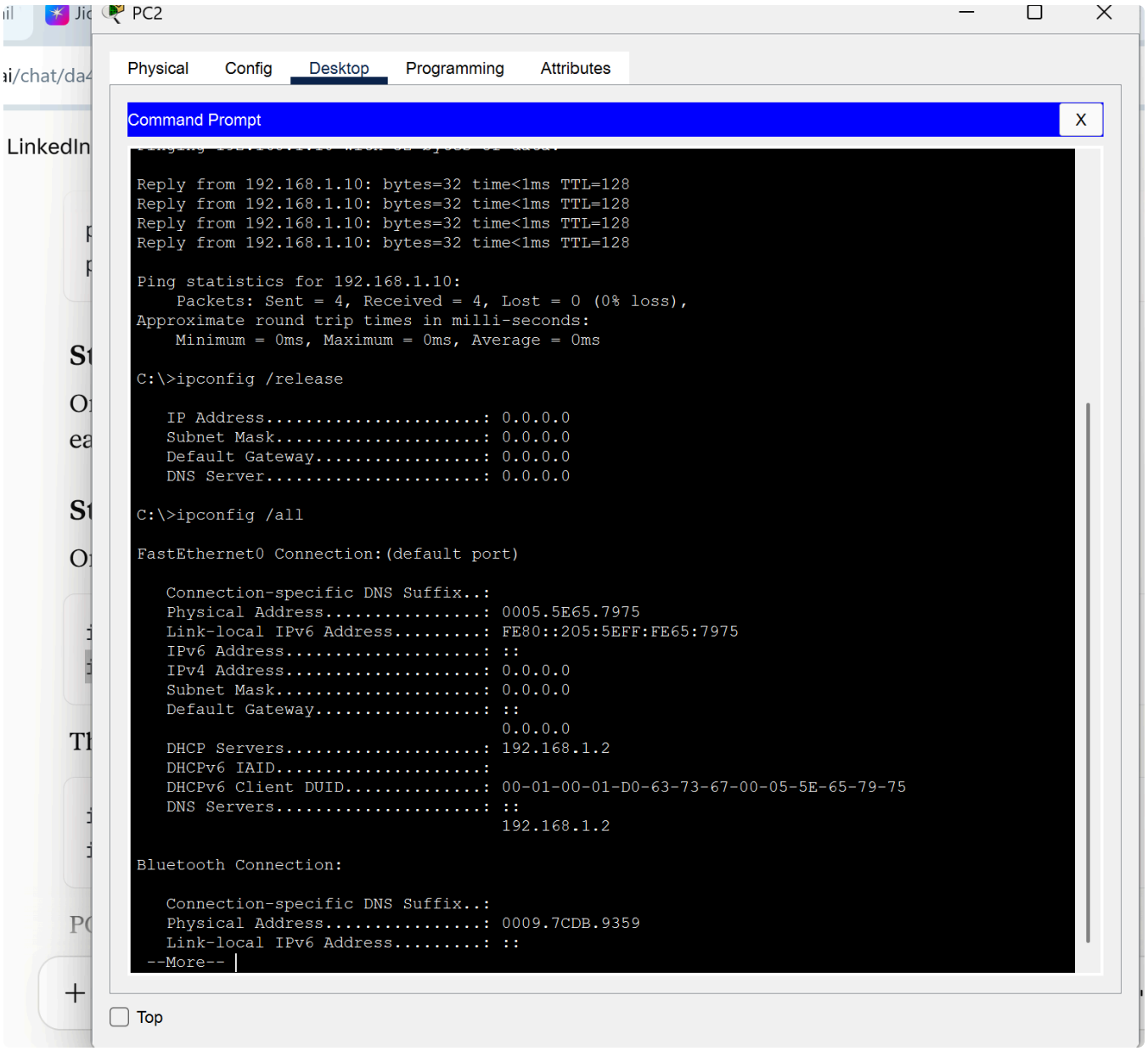


Figure 1: PC2 after ipconfig /release, with every address field reset to 0.0.0.0.

5.2 Renew: getting a new lease

Running `ipconfig /renew` on PC2 started a new DORA exchange, and the server answered with a full configuration from the pool:

Setting	Value received
IP address	192.168.1.12
Subnet mask	255.255.255.0

Setting	Value received
Default gateway	192.168.1.1
DNS server	192.168.1.2
DHCP server	192.168.1.2

The address 192.168.1.12 is inside the pool that starts at .10, and the gateway and DNS values match what was configured on the server.

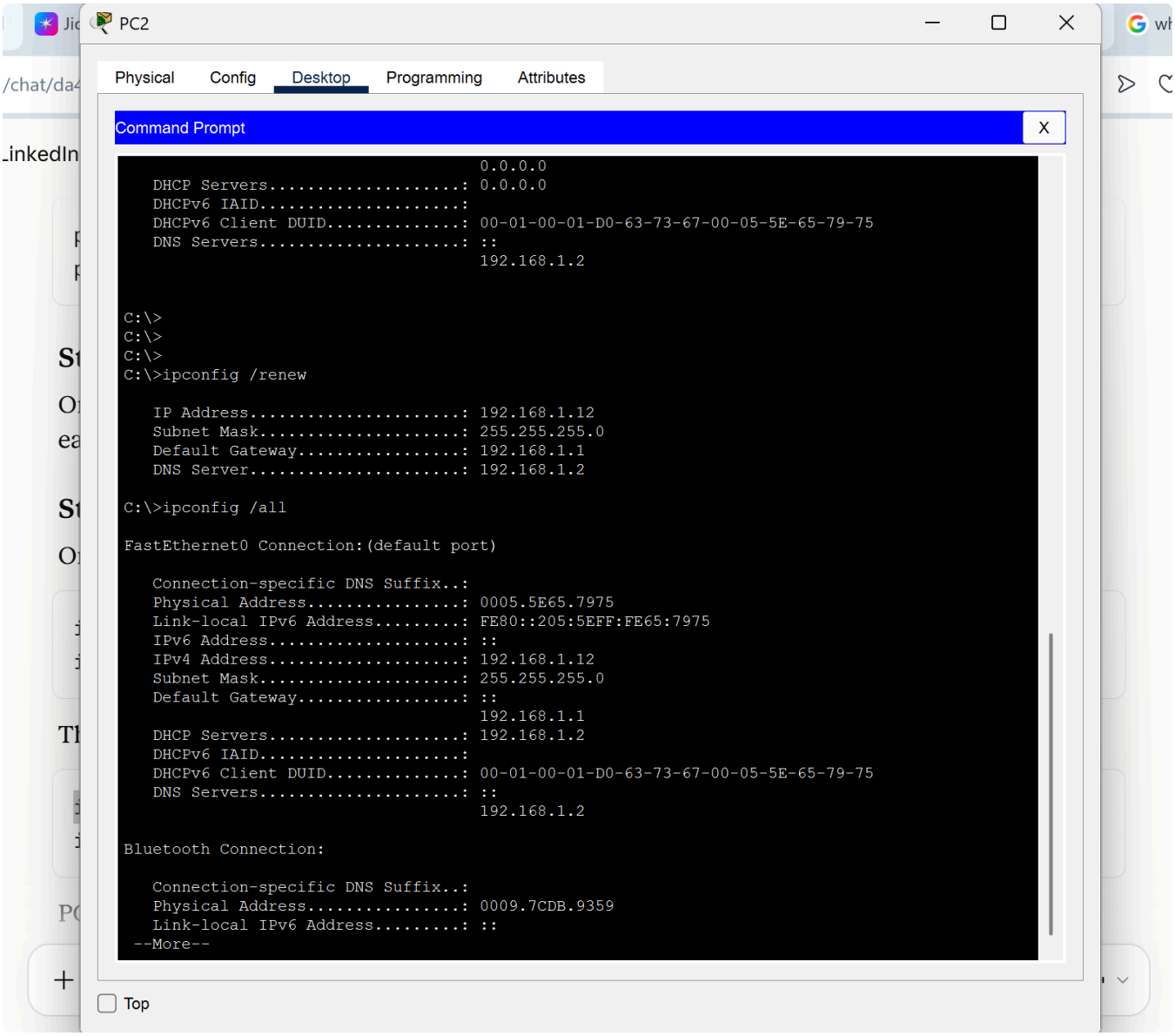


Figure 2: PC2 after ipconfig /renew, then ipconfig /all confirming 192.168.1.12 from DHCP server 192.168.1.2.

5.3 Connectivity test

PC2 pinged 192.168.10.1 first and every request timed out (100% loss). That address is on a different network (192.168.10.0/24) from the DHCP subnet (192.168.1.0/24), so there is nothing to answer it; it was probably a mistyped 192.168.1.1. PC2 then pinged 192.168.1.10 and got 4 of 4 replies (0% loss, TTL 128), which shows the DHCP-assigned address works for normal communication on the LAN.

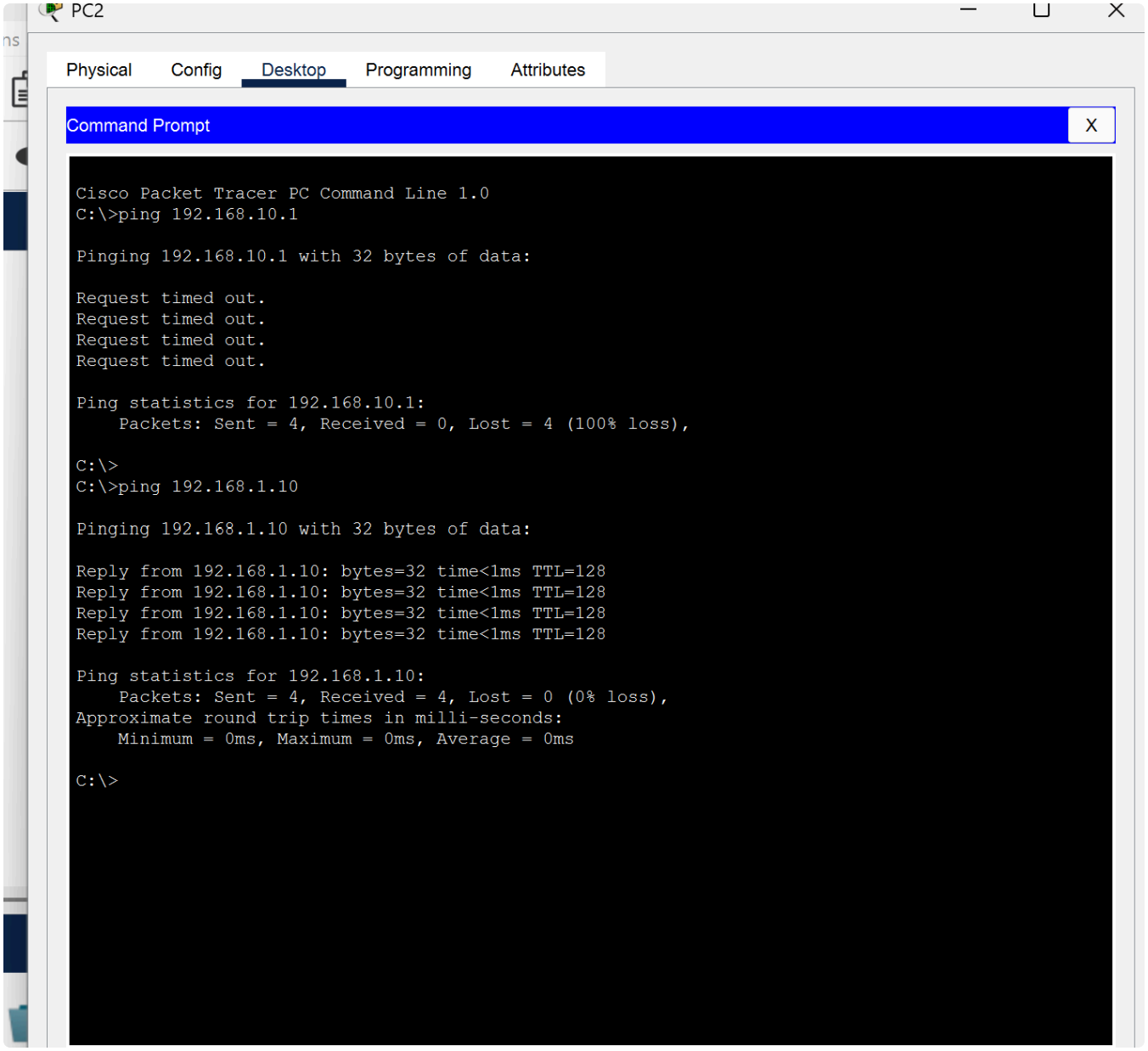


Figure 3: PC2 pinging 192.168.10.1 (fails, wrong network) and 192.168.1.10 (succeeds).

6. Lease Table

The server records each lease with the client's hardware (MAC) address and the IP assigned to it. The row below comes from the screenshots; the rows for the other clients come from the server's lease table (Server > Services > DHCP, or `show ip dhcp binding` if a router is the server).

Client	MAC address	IP address	DHCP server
PC2	0005.5E65.7975	192.168.1.12	192.168.1.2

7. Explanation

What DHCP does. DHCP (Dynamic Host Configuration Protocol) automatically gives devices the network settings they need: an IP address, subnet mask, default gateway and DNS server. Without it, every device would have to be configured by hand, which is slow and invites mistakes such as duplicate addresses.

The scope (pool). The scope is the range of addresses the server may hand out, here starting at 192.168.1.10 and covering up to 50 addresses. Anything outside it, like the server's own .2 and the gateway's .1, is never given to a client.

A lease is temporary. The address is loaned to the client, not given permanently. While the lease is active, the server will not hand that address to anyone else. Clients try to renew automatically at about half the lease time. If a client releases its lease (as in Figure 1) or the lease expires and is not renewed, the address returns to the pool and can be given to another device.

Why renew often gives the same address. The server remembers which MAC address had which IP, so a returning client usually gets the same address again, as long as it is still free.

Pool size is a limit. If every address in the pool is leased, the next client gets no address at all, and Windows falls back to a 169.254.x.x self-assigned address.

8. Results

Test	Outcome
Client receives address automatically	Passed (192.168.1.12)
Address is inside the pool, with correct mask, gateway and DNS	Passed
DHCP server identified as 192.168.1.2	Passed
Release returns all fields to 0.0.0.0	Passed
Renew obtains a fresh lease	Passed
Ping to a host on the LAN	Passed (0% loss)
Ping to an address on a different network	Failed as expected

9. Conclusion

The DHCP server handed out complete network settings from its pool with no manual configuration on the client. Release and renew showed the lease lifecycle: the address is returned and reset to 0.0.0.0, then a new DORA exchange issues a fresh lease from the same pool. DHCP removes the manual work and the risk of duplicate addresses in a network of any size.

10. Screenshots Still Needed for a Complete Submission

1. The server's **Services > DHCP** page showing the pool settings
2. The server's lease table (Show Leases, or `show ip dhcp binding` on a router), listing all three clients
3. `ipconfig /all` on PC0 and PC1 showing their DHCP-assigned addresses
4. A **DORA capture**: in Packet Tracer Simulation mode with the DHCP filter on, run `ipconfig /renew` on a client and capture the Discover, Offer, Request and Acknowledge packets. Wireshark is not available inside Packet Tracer; it applies only to a VM lab.